Life-cycle model with search frictions and skills

March 30, 2026

1 Model

An individual is characterized by assets $a \geq 0$, employment status $\ell \in \{0, 1\}$, skill level $g \in \{\bar{g}_1, \ldots, \bar{g}_{n_g}\}$, age $j \in \{1, \ldots, J\}$, and a fixed education type $i \in \mathcal{I} \equiv \{\text{low}, \text{high}\}$. Here $\ell = 0$ denotes unemployment and $\ell = 1$ denotes employment. Retirement begins at age J_R.

Education affects the environment through two type-specific objects: $\phi_i > 0$, an income shifter, and $\sigma_i \in (0, 1)$, the job-separation probability while employed. We assume $\sigma_{\text{low}} > \sigma_{\text{high}}$, so higher-education individuals face a lower risk of job loss.

At each age, the individual chooses next-period assets $a' \geq 0$ and search effort $s \geq 0$. Search effort matters only while the agent is unemployed and of working age, that is, when $\ell = 0$ and $j < J_R$.

For all ages $j \leq J$, the Bellman equation is

$$V_i(a, \ell, g, j) = \max_{a' \geq 0,\, s \geq 0} \left\{ u(c) - \kappa(s) + \beta \sum_{\ell' \in \{0,1\}} \sum_{g' \in \mathcal{G}} V_i(a', \ell', g', j+1)\, \pi_\ell(\ell, \ell' \mid s, i)\, \pi_g(g, g' \mid \ell, j) \right\},$$

where $\mathcal{G} = \{\bar{g}_1, \ldots, \bar{g}_{n_g}\}$, with terminal condition

$$V_i(a, \ell, g, J+1) = 0.$$

Associated policy functions for next-period assets and search effort are

$$Pol_{a'}(a, \ell, g, j, i), \qquad Pol_s(a, \ell, g, j, i).$$

The period budget constraint is

$$\begin{cases} c + a' = (1+r)a + \phi_i \left[w\, g\, \mathbb{I}_{\{\ell=1\}} + b\, \mathbb{I}_{\{\ell=0\}} \right], & j < J_R, \\ c + a' = (1+r)a + \phi_i\, pen, & j \geq J_R. \end{cases}$$

Thus, at working ages labor income equals $\phi_i w g$ when employed and $\phi_i b$ when unemployed, while retirees receive pension income $\phi_i pen$. Feasible choices must satisfy $c > 0$.

Period utility is

$$u(c) = \frac{c^{1-\gamma}}{1-\gamma}, \qquad \kappa(s) = B_s s.$$

Since $\kappa(s)$ is strictly increasing, the optimal choice in all states in which search brings no benefit is the lowest feasible effort, namely $s = 0$. In particular, search is irrelevant while employed and after retirement, when employment status no longer affects payoffs. Likewise, after retirement the payoff function no longer depends on ℓ or g, but we still keep those variables in the state vector to maintain a uniform recursive formulation and to match the Matlab implementation.

Employment transitions at working ages depend on search effort:

$$\pi_\ell(\ell, \ell' \mid s, i) = \begin{bmatrix} & \ell' = 0 & \ell' = 1 \\ \ell = 0 & 1 - \pi^s(s) & \pi^s(s) \\ \ell = 1 & \sigma_i & 1 - \sigma_i \end{bmatrix}.$$

An unemployed worker finds a job with probability $\pi^s(s)$, while an employed worker loses the job with probability σ_i. This matrix is well defined provided $\pi^s(s) \in [0, 1]$.

Skill transitions depend on current employment status. A simple three-state illustration is

$$\pi_g(g, g' \mid \ell = 0, j) = \begin{bmatrix} & g' = \bar{g}_1 & g' = \bar{g}_2 & g' = \bar{g}_3 \\ g = \bar{g}_1 & 1 & 0 & 0 \\ g = \bar{g}_2 & 1 - p_0 & p_0 & 0 \\ g = \bar{g}_3 & 0 & 1 - p_0 & p_0 \end{bmatrix},$$

and

$$\pi_g(g, g' \mid \ell = 1, j) = \begin{bmatrix} & g' = \bar{g}_1 & g' = \bar{g}_2 & g' = \bar{g}_3 \\ g = \bar{g}_1 & p_1^j & 1 - p_1^j & 0 \\ g = \bar{g}_2 & 0 & p_1^j & 1 - p_1^j \\ g = \bar{g}_3 & 0 & 0 & 1 \end{bmatrix}.$$

Under this ladder structure, unemployment weakly depreciates skills, while employment weakly improves them. To make the matrices stochastic, one needs $p_0 \in [0, 1]$ and $p_1^j \in [0, 1]$ for every age j. The age index in p_1^j allows skill accumulation to vary over the life cycle, for example by making it stronger at younger ages.

1.1 Distribution

Let ϖ_i denote the fixed population share of permanent education type $i \in \mathcal{I}$, with $\sum_{i \in \mathcal{I}} \varpi_i = 1$. Let $m_j \geq 0$ denote the cross-sectional age-mass profile, with $\sum_{j=1}^{J} m_j = 1$. In the code this object is stored as $m_j = Params.mewj(j)$. The current application sets $m_j = 1/J$ for all j, but the formulation allows for a general age distribution, for example one generated by survival risk. Let $\tilde{\mu}(a, \ell, g, j, i)$ denote the distribution over (a, ℓ, g, i) within age j, before applying the cross-sectional age weights, and let $\mu(a, \ell, g, j, i)$ denote the joint cross-sectional distribution. These objects are related by

$$\mu(a, \ell, g, j, i) = m_j \, \tilde{\mu}(a, \ell, g, j, i).$$

Because permanent type is fixed over the life cycle, both distributions keep track of i explicitly rather than integrating it out. After retirement, the distribution still carries ℓ and g, even though they are payoff-irrelevant, in order to preserve the same state-space representation at all ages.

Given these policy functions, the within-age distribution evolves according to

$$\begin{aligned} \tilde{\mu}(a', \ell', g', j+1, i) \quad = \quad & \sum_{a, \ell, g} \tilde{\mu}(a, \ell, g, j, i) \, \mathbb{I}\left\{a' = Pol_{a'}(a, \ell, g, j, i)\right\} \\ & \times \pi_\ell(\ell, \ell' \mid Pol_s(a, \ell, g, j, i), i) \, \pi_g(g, g' \mid \ell, j), \end{aligned}$$

for $j = 1, \ldots, J-1$. Equivalently, the cross-sectional distribution satisfies

$$\begin{aligned} \mu(a', \ell', g', j+1, i) \quad = \quad & \frac{m_{j+1}}{m_j} \sum_{a, \ell, g} \mu(a, \ell, g, j, i) \, \mathbb{I}\left\{a' = Pol_{a'}(a, \ell, g, j, i)\right\} \\ & \times \pi_\ell(\ell, \ell' \mid Pol_s(a, \ell, g, j, i), i) \, \pi_g(g, g' \mid \ell, j), \end{aligned}$$

whenever $m_j > 0$.

Let $g_0 \in \mathcal{G}$ denote the median skill level. The initial within-age distribution at age $j = 1$ is given by

$$\tilde{\mu}(a, \ell, g, 1, i) = \varpi_i \, \mathbb{I}\{a = 0\} \, \mathbb{I}\{\ell = 0\} \, \mathbb{I}\{g = g_0\},$$

so all newborn agents start with zero assets, unemployed, and at the median skill level g_0, while permanent types are distributed according to ϖ_i. Therefore the age-1 cross-sectional distribution is

$$\mu(a, \ell, g, 1, i) = m_1 \, \varpi_i \, \mathbb{I}\{a = 0\} \, \mathbb{I}\{\ell = 0\} \, \mathbb{I}\{g = g_0\}.$$

1.2 Life-cycle profiles

Consider a generic variable $x = x(a, \ell, g, j, i)$ defined on the individual state space. Examples include assets, search effort, and employment status. Using the cross-sectional distribution $\mu(a, \ell, g, j, i)$, the average of x conditional on age j is

$$\mathbb{E}[x \mid j] = \frac{\sum_{i \in \mathcal{I}} \sum_{a,\ell,g} x(a, \ell, g, j, i) \, \mu(a, \ell, g, j, i)}{\sum_{i \in \mathcal{I}} \sum_{a,\ell,g} \mu(a, \ell, g, j, i)}.$$

Since μ is the joint cross-sectional distribution and the age mass at age j is m_j, the denominator equals m_j. Hence one can also write

$$\mathbb{E}[x \mid j] = \frac{1}{m_j} \sum_{i \in \mathcal{I}} \sum_{a,\ell,g} x(a, \ell, g, j, i) \, \mu(a, \ell, g, j, i).$$

If instead one wants the average conditional on both age j and permanent type i, then

$$\mathbb{E}[x \mid j, i] = \frac{\sum_{a,\ell,g} x(a, \ell, g, j, i) \, \mu(a, \ell, g, j, i)}{\sum_{a,\ell,g} \mu(a, \ell, g, j, i)}.$$

Because permanent type shares are fixed over the life cycle, the denominator is $m_j \varpi_i$, so equivalently

$$\mathbb{E}[x \mid j, i] = \frac{1}{m_j \varpi_i} \sum_{a,\ell,g} x(a, \ell, g, j, i) \, \mu(a, \ell, g, j, i).$$

If one wants the average of x conditional on age j and employment status $\ell = 0$, then

$$\mathbb{E}[x \mid j, \ell = 0] := \frac{\sum_{i \in \mathcal{I}} \sum_{a,g} x(a, 0, g, j, i) \, \mu(a, 0, g, j, i)}{\sum_{i \in \mathcal{I}} \sum_{a,g} \mu(a, 0, g, j, i)}.$$

Similarly, conditional on age j and employment status $\ell = 1$,

$$\mathbb{E}[x \mid j, \ell = 1] := \frac{\sum_{i \in \mathcal{I}} \sum_{a,g} x(a, 1, g, j, i) \, \mu(a, 1, g, j, i)}{\sum_{i \in \mathcal{I}} \sum_{a,g} \mu(a, 1, g, j, i)}.$$

More compactly, for any $\bar{\ell} \in \{0, 1\}$,

$$\mathbb{E}[x \mid j, \ell = \bar{\ell}] := \frac{\sum_{i \in \mathcal{I}} \sum_{a,g} x(a, \bar{\ell}, g, j, i) \, \mu(a, \bar{\ell}, g, j, i)}{\sum_{i \in \mathcal{I}} \sum_{a,g} \mu(a, \bar{\ell}, g, j, i)}.$$

Other moments by age, or jointly by age and permanent type, can be computed in exactly the same way by replacing x with the relevant function of the state variables.

Implementation note. In the VFI Toolkit, this problem is implemented as a finite-horizon model with semi-exogenous states and no purely exogenous Markov state z. We define the semi-exogenous state as $semiz = (\ell, g)$. The reason is that the employment transition depends on the decision variable $d = s$, while skill transitions depend on current employment status ℓ. Hence ℓ is directly semi-exogenous, and g is naturally treated as a second semi-exogenous state.

Variable	Description	Toolkit name
s	Search effort	d
a'	Next-period endogenous state	$aprime$
a	Current endogenous state	a
ℓ	Employment state	$semiz1$
g	Skill level	$semiz2$

In that notation, the relevant objects are the semi-exogenous grid $semiz_grid_J$ and the transition array $\pi_{semiz,J}$. In full generality these can be entered as multidimensional arrays:

$$semiz_grid_J(semiz, j, ptype), \qquad \pi_{semiz,J}(semiz, semiz', d, j, ptype).$$

If the semi-exogenous grid is the same for all ages and permanent types, however, it can be entered in stacked form as a column vector $semiz_grid_J(semiz)$ rather than as a joint grid. See the VFI Toolkit documentation or code for further details on this notation and data layout.